

Flow of the Program — Flowcharts & Pseudocode

Designing and Visualizing Program Logic Before Writing Java Code

Introduction to Program Flow

Program flow refers to the order in which individual statements, instructions, or function calls are executed in a program. Before writing code in Java, developers design and visualize this flow using **Flowcharts** and **Pseudocode** to prevent logic errors.

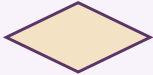
What Is a Flowchart?

Explanation: A flowchart is a diagrammatic or visual representation of an algorithm or a step-by-step process. It uses standard geometric shapes connected by arrows to show how data and control move from start to finish.

Real-World Example: A traffic navigation sign board showing: "If right lane is closed → Turn Left → Otherwise Continue Straight".

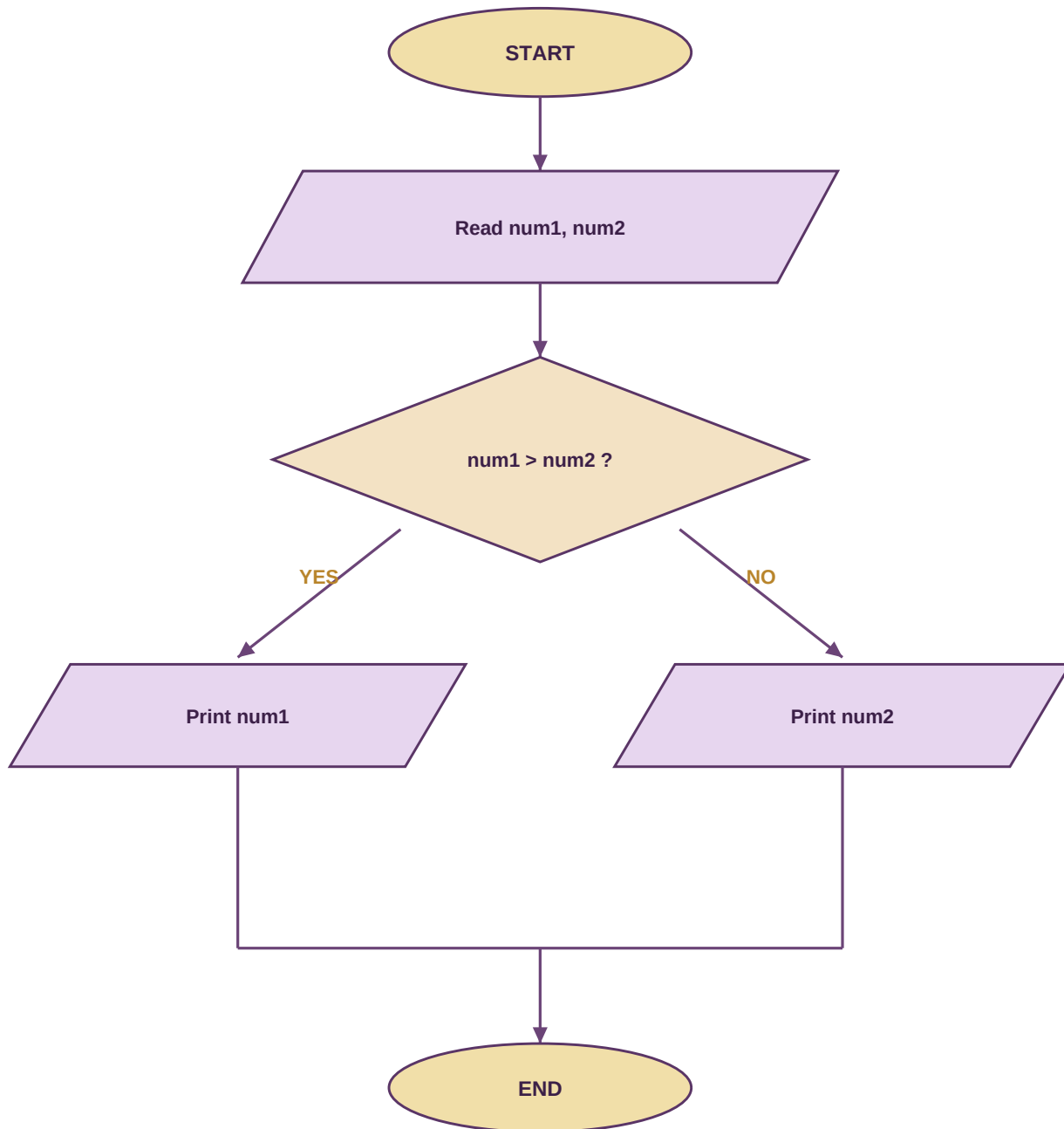
Programming Example: Checking if a student passed an exam based on whether their marks are ≥ 40 .

Important Flowchart Components

Shape	Symbol	What It Does
	Terminal (Start / End)	Marks the beginning or end of the program flow.
	Input / Output	Reads or prints data (e.g., Read num, Print sum).
	Process	Arithmetic operations, variable assignments (e.g., $c = a + b$).
	Decision	Condition check returning True (Yes) or False (No).
	Flowline (Arrow)	Shows the direction of the program flow.
	Connector	Connects different flow sections on a complex chart.

Flowchart Example — Find the Largest of Two Numbers

1. Visual Flowchart Diagram



The decision box checks num1 > num2. Both branches print a result and merge back into a single path before the flow reaches END.

2. Corresponding Java Implementation

JAVA

```
import java.util.Scanner;

public class LargestOfTwo {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int num1 = sc.nextInt();
        int num2 = sc.nextInt();

        if (num1 > num2) {
            System.out.println("Largest: " + num1);
        } else {
            System.out.println("Largest: " + num2);
        }

        sc.close();
    }
}
```

Pseudocode: Definition, Uses, and Rules

1. What Is Pseudocode?

Explanation: Pseudocode is an informal, high-level, plain-English description of a computer program's logic. It does not follow strict programming language syntax (like semicolons or class declarations), making it easily readable by humans.

Real-World Example: Writing a quick outline of a speech using bullet points before drafting the full formal speech.

2. Uses of Pseudocode

Use	Description
Bridge between concept and code	Helps translate abstract thoughts into structured programming logic.
Language-independent	The same pseudocode can be converted into Java, Python, C++, or any other language.
Easy debugging	Logic errors can be spotted and fixed before typing complex code.

3. Pseudocode Example: Check Even or Odd

PSEUDOCODE

```
BEGIN
    READ number
    IF number MOD 2 == 0 THEN
        PRINT "Even Number"
    ELSE
        PRINT "Odd Number"
    ENDIF
END
```

Corresponding Java Implementation

JAVA

```
public class EvenOddDemo {
    public static void main(String[] args) {
        int number = 7;

        if (number % 2 == 0) {
            System.out.println("Even Number");
        } else {
            System.out.println("Odd Number");
        }
    }
}
```

Practice Questions for Students (Unsolved)

Flowchart Questions

#	Question
1	Draw a flowchart to find whether a given year is a Leap Year or not.
2	Draw a flowchart to print the first N natural numbers using a loop.
3	Draw a flowchart to calculate the simple interest given Principal (P), Time (T), and Rate of Interest (R).

Pseudocode Questions

#	Question
1	Write pseudocode to accept three numbers and determine the smallest among them.
2	Write pseudocode to calculate the factorial of a positive integer N (e.g., $5! = 5 \times 4 \times 3 \times 2 \times 1$).
3	Write pseudocode to check whether a given number is Prime or Not Prime.